



Voice MFA System

Technical Project Report

Multi-Factor Authentication Using Voice Biometrics

Prepared by:

Tibbeche abed el hassib Ammar

Ouaret wail

Abdenmour khaled

Benyoub khalil

Supervisor:

Pr. Lounis karim

Academic Year:

2025–2026

Secure Multi-Factor Authentication Using Voice Biometrics
National Higher School of Artificial Intelligence

Contents

Abstract	3
1 Introduction	4
2 Project Objectives	5
3 System Overview	6
3.1 Frontend	6
3.2 Backend	6
3.3 Database	6
4 Voice Verification System	8
4.1 Enrollment Phase	8
4.2 Verification Phase	8
4.3 Cosine Similarity	8
5 Security Mechanisms	10
5.1 Password Hashing	10
5.2 JWT Authentication	10
5.3 Challenge Phrases	10
5.4 TOTP Verification	10
6 Database Design	11
6.1 Users Table	11

6.2	Voice Profiles Table	11
6.3	Audit Logs Table	11
7	System Workflow	12
7.1	Authentication Flow	12
8	Security Analysis	13
8.1	Strengths	13
8.2	Limitations	13
9	Future Improvements	14
10	Conclusion	15
	References	16

Abstract

This project presents the design and implementation of a Voice Multi-Factor Authentication (Voice MFA) system that combines password authentication, voice biometrics, and optional Time-Based One-Time Password (TOTP) verification.

The system demonstrates how biometric authentication can strengthen traditional login mechanisms by introducing speaker verification as an additional security factor.

The project contains two implementations:

- A local Python prototype
- A full-stack web application using FastAPI and React

Voice verification is implemented using Resemblyzer speaker embeddings and cosine similarity comparison. Additional security mechanisms include bcrypt password hashing, JWT session management, Redis challenge storage, backup codes, and audit logging.

Chapter 1

Introduction

Authentication systems traditionally rely on passwords. However, passwords alone are vulnerable to phishing, leaks, brute-force attacks, and password reuse.

Multi-Factor Authentication (MFA) improves security by combining multiple authentication factors:

- Something the user knows (password)
- Something the user is (voice biometrics)
- Something the user has (TOTP authenticator)

This project implements a Voice MFA architecture integrating these layers into a modern authentication workflow.

Chapter 2

Project Objectives

The main objectives of this project are:

- Build a secure MFA authentication system
- Implement voice biometric verification
- Integrate TOTP authentication
- Create browser-based audio recording
- Store and compare voice embeddings
- Implement JWT session handling
- Add audit logging and account lockout protection

Chapter 3

System Overview

The Voice MFA system contains several components.

3.1 Frontend

- React + Vite
- Browser audio recording
- User dashboard
- MFA verification pages

3.2 Backend

- FastAPI REST API
- SQLAlchemy ORM
- JWT authentication
- Voice verification services

3.3 Database

- PostgreSQL or SQLite
- User accounts
- Voice embeddings

- Audit logs

Chapter 4

Voice Verification System

4.1 Enrollment Phase

During enrollment:

1. The user records several voice samples
2. Audio is converted into NumPy arrays
3. Resemblyzer extracts 256-dimensional embeddings
4. Embeddings are stored in the database

4.2 Verification Phase

During authentication:

1. The user receives a challenge phrase
2. A new voice sample is recorded
3. The backend extracts a new embedding
4. Cosine similarity compares embeddings

4.3 Cosine Similarity

The similarity between embeddings is calculated using:

$$\text{similarity}(a, b) = \frac{a \cdot b}{||a|| ||b||}$$

A value close to 1 indicates similar speakers.

Chapter 5

Security Mechanisms

5.1 Password Hashing

Passwords are hashed using bcrypt before storage.

5.2 JWT Authentication

The system uses:

- Pre-authentication tokens
- Full-session tokens

5.3 Challenge Phrases

Random phrases reduce replay attacks.

Example:

glacier raven ember 742

5.4 TOTP Verification

The system supports Time-Based One-Time Password authentication using authenticator applications.

Chapter 6

Database Design

6.1 Users Table

Stores:

- Username
- Email
- Password hash
- TOTP status
- Voice enrollment status

6.2 Voice Profiles Table

Stores serialized voice embeddings.

6.3 Audit Logs Table

Stores security events such as:

- Login attempts
- Voice verification
- Enrollment
- Lockout events

Chapter 7

System Workflow

7.1 Authentication Flow

1. User enters username and password
2. Backend validates credentials
3. Backend issues pre-auth token
4. User records voice sample
5. Backend verifies speaker identity
6. Optional TOTP verification occurs
7. Backend issues full-session token

Chapter 8

Security Analysis

8.1 Strengths

- Multi-factor authentication
- Voice embeddings
- bcrypt password hashing
- Redis challenge expiration
- Audit logging
- Account lockout protection

8.2 Limitations

- No anti-spoofing detection
- No deepfake detection
- No spoken phrase verification in web version

Chapter 9

Future Improvements

Possible improvements include:

- Anti-spoofing systems
- Deepfake detection
- HTTPS deployment
- Rate limiting
- Session management improvements

Chapter 10

Conclusion

Voice MFA demonstrates how voice biometrics can strengthen traditional authentication systems.

The project combines:

- Web development
- Audio processing
- Machine learning
- Cybersecurity
- Database systems

The use of speaker embeddings, cosine similarity, JWT separation, and TOTP verification creates a realistic MFA architecture suitable for educational and research purposes.

References

1. Resemblyzer Documentation
2. FastAPI Documentation
3. SQLAlchemy Documentation
4. JWT RFC 7519
5. TOTP RFC 6238
6. Redis Documentation
7. bcrypt Documentation
8. Docker Documentation